NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2024-2025

IE2106 – ENGINEERING MATHEMATICS I

November / December 2024 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 13 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.
6. A list of useful formulae is given in Appendix A on page 5, followed by Statistical Tables on pages 7 - 11, and the Laplace Transform Table on pages 12-13.

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1. (a) In the central kitchen of a buffet caterer, historically 20% of spring rolls produced have to be discarded because their skin is torn during the frying process.

   (i) A worker in the central kitchen randomly picked 10 spring rolls for packing, and he got the impression that there were at most 5 spring rolls with torn skin. What is the probability that there are at least 2 spring rolls in the sample that have to be discarded?

   (ii) The central kitchen manager is trying to improve the production efficiency by focusing on p, the proportion of spring rolls produced in the central kitchen that has to be discarded. Find the range of p (in percentage) so that the probability of having at least 1 spring roll with torn skin in a random sample of 100 spring rolls is at most 10%.

   (9 Marks)

Note: Question No. 1 continues on page 2.

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(b) A bakery that is famous for curry puffs has put in place the following production standards:
(S1) the mean weight of filling in a curry puff should be 25 g, and
(S2) the standard deviation of the weights of fillings should be below 2 g.
To check whether the standards are met, a random sample of 41 curry puffs is taken and it is measured that the weights of fillings have a mean of 24.7 g and a standard deviation of 1.9 g. Conduct **two** hypothesis tests at the 5% level of significance to see if the production standards (S1) and (S2) are upheld. State any assumptions needed for each test. How would you advise the bakery to improve their operation?

(16 Marks)

2. (a) Let X be the number of car seats in a car with registered address in Jurong. The possible values of X are 2, 4, 5, 7 and

$$P(X=2)=a, \quad P(X=4)=0.25, \quad P(X=5)=b, \quad P(X=7)=0.3,$$

where a and b are constants. It is known that the expected value of X is 5.2. Find a and b. A car park attendant found that a car was illegally parked in the wheel chair lot of a HDB car park at Jurong West. The car appeared to be bigger than a two-seater and it is most likely owned by a resident. What is the probability that the car has at most 5 seats?

(7 Marks)

(b) Consider the equation

$$2\sin^2 x - \frac{1}{2}x^2 + \frac{5}{2} = 0, \qquad x \in \left[\frac{\pi}{2}, \pi\right]. \qquad (*)$$

(i) Show that the equation (*) has at least one solution in $[\pi/2, \pi]$.

(ii) To solve equation (*) by the fixed point iteration scheme $x_{n+1} = g(x_n)$, $n = 0, 1, \cdots$, what condition(s) is (are) needed to guarantee convergence of the scheme?

(iii) The equation (*) can be written as

$$x = (4\sin^2 x + 5)^{1/2} \equiv g_1(x) \qquad \text{or} \qquad x = \frac{4\sin^2 x + 5}{x} \equiv g_2(x).$$

Investigate which function ($g_1(x)$ or $g_2(x)$) will give a fixed point iteration scheme that converges to the solution of (*). Hence, carry out the computation to obtain an approximate solution of (*). Set up a criterion to ensure that the approximate solution obtained is correct to 2 decimal places. Start with $x_0 = 2.53$.

(18 Marks)

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3. (a) (i) Find the Laplace transform of the following function

$$f(t) = t[u(t) - u(t-3)]$$

where $u(t)$ is the unit step function.

(ii) Find the inverse Laplace transform of the following function

$$G(s) = \frac{0.4}{s^2(s+0.8)}e^{-s}$$

(10 Marks)

(b) The response of a linear system to a unit step input is described by the following equation:

$$y(t) = (1 - e^{-2t})u(t)$$

where $u(t)$ is the unit step function and $y(t)$ is the response. Find the transfer function of the system, and use the transfer function to find the response of the linear system to the following input:

$$x(t) = \begin{cases} 0 & t < 0 \\ t & 0 \le t \le 1 \\ 1 & t > 1 \end{cases}$$

(15 Marks)

4. (a) Consider the Partial Differential Equation (PDE)

$$U_t(t,x) = c^2 U_{xx}(t,x)$$

where c is a constant.

(i) What is the order of this PDE? Is the PDE linear or nonlinear, variant or invariant, homogeneous or non-homogeneous?

(ii) Show that

$$U(t,x) = e^{-\pi^2 t}\cos(25x)$$

is a solution of the PDE in part 4a(i). Give also the value of c^2.

(12 Marks)

Note: Question No. 4 continues on page 4.

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(b) By the method of separation of variables, find the general solution of the Partial Differential Equation

$$U_{tt}(t,x) = 4U_{xx}(t,x)$$

where

$$U(t,0) = U(t,\pi) = 0 \text{ for all } t, \text{ and } U(0,x) = \sin(2x).$$

(13 Marks)

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Appendix A

Total Probability of A

$$P(A) = \sum_{j=1}^{k} P(A \mid B_j) P(B_j)$$

Bayes Theorem

$$P(B_r \mid A) = \frac{P(A \mid B_r) P(B_r)}{\sum_{j=1}^{k} P(A \mid B_j) P(B_j)}$$

Poisson Distribution

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Fixed-Point Iteration Method

$$x_{n+1} = g(x_n), \quad n = 0, 1, 2, \ldots$$

Euler Method

$$y_{n+1} = y_n + hf(x_n, y_n)$$

Improved Euler Method (Heun's Method)

$$a = hf(x_n, y_n)$$
$$b = hf(x_{n+1}, y_n + a)$$
$$y_{n+1} = y_n + \tfrac{1}{2}(a + b)$$

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Appendix A (Continued)

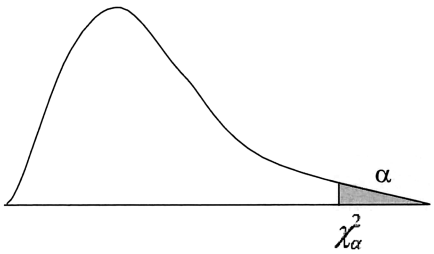
Runge-Kutta Method

$$a = h\,f(x_n, y_n)$$
$$b = h\,f(x_n + \tfrac{1}{2}h,\ y_n + \tfrac{1}{2}a)$$
$$c = h\,f(x_n + \tfrac{1}{2}h,\ y_n + \tfrac{1}{2}b)$$
$$d = h\,f(x_n + h,\ y_n + c)$$
$$y_{n+1} = y_n + \tfrac{1}{6}(a + 2b + 2c + d)$$

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Statistical Table - 1

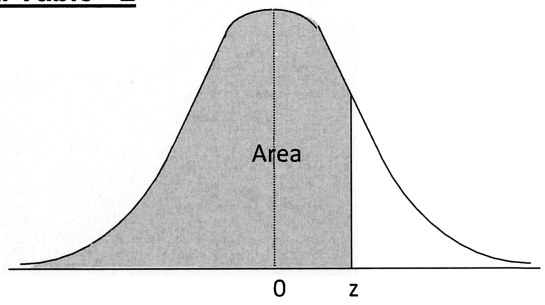
Percentage Points of χ^2 Distribution



d.f. \ α	.99	.975	.95	.90	.50	.10	.05	.025	.01
1	.0002	.001	.004	.02	.45	2.71	3.84	5.02	6.63
2	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21
3	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34
4	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28
5	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09
6	.87	1.24	1.64	2.20	5.35	10.64	12.59	14.45	16.81
7	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48
8	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09
9	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67
10	2.56	3.24	3.94	4.87	9.34	15.99	18.31	20.48	23.21
11	3.05	3.81	4.57	5.58	10.34	17.28	19.68	21.92	24.72
12	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22
13	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69
14	4.66	5.62	6.57	7.79	13.34	21.06	23.68	26.12	29.14
15	5.23	6.26	7.26	8.55	14.34	22.31	25.00	27.49	30.58
16	5.81	6.90	7.96	9.31	15.34	23.54	26.30	28.85	32.00
17	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41
18	7.01	8.23	9.39	10.86	17.34	25.99	28.87	31.53	34.81
19	7.63	8.90	10.12	11.65	18.34	27.20	30.14	32.85	36.19
20	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57
21	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93
22	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29
23	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64
24	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98
25	11.52	13.11	14.61	16.47	24.34	34.38	37.65	40.65	44.31
26	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64
27	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96
28	13.56	15.30	16.93	18.94	27.34	37.92	41.34	44.46	48.28
29	14.26	16.04	17.71	19.77	28.34	39.09	42.56	45.72	49.59
30	14.95	16.78	18.49	20.60	29.34	40.26	43.77	46.98	50.89
40	22.16	24.42	26.51	29.05	39.34	51.81	55.76	59.34	63.69
50	29.71	32.35	34.76	37.69	49.33	63.17	67.50	71.42	76.15
60	37.48	40.47	43.19	46.46	59.33	74.40	79.08	83.30	88.38
70	45.44	48.75	51.74	55.33	69.33	85.53	90.53	95.02	100.43
80	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33
90	61.75	65.64	69.13	73.29	89.33	107.57	113.15	118.14	124.12
100	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81

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Statistical Table - 2



Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Note: Table 2 continues on page 9.

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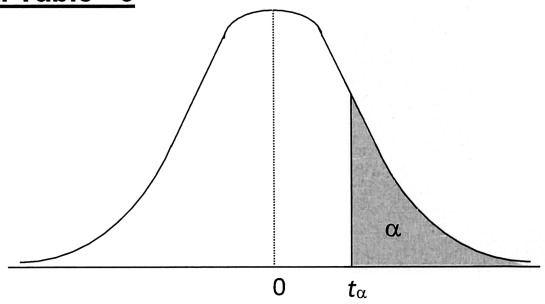
Statistical Table – 2 (Continued)

Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9278	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

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Statistical Table - 3



Critical Values of the *t*-Distribution

	α						
ν	0.40	0.30	0.20	0.15	0.10	0.05	0.025
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179
13	0.259	0.537	0.870	1.079	1.350	1.771	2.160
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960

Note: Table 3 continues on page 11.

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Statistical Table - 3 (Continued)

Critical Values of the *t*-Distribution

	α						
ν	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	15.895	21.205	31.821	42.434	63.657	127.322	636.590
2	4.849	5.643	6.965	8.073	9.925	14.089	31.598
3	3.482	3.896	4.541	5.047	5.841	7.453	12.924
4	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	2.612	2.829	3.143	3.372	3.707	4.317	5.959
7	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	2.398	2.574	2.821	2.998	3.250	3.690	4.781
10	2.359	2.527	2.764	2.932	3.169	3.581	4.587
11	2.328	2.491	2.718	2.879	3.106	3.497	4.437
12	2.303	2.461	2.681	2.836	3.055	3.428	4.318
13	2.282	2.436	2.650	2.801	3.012	3.372	4.221
14	2.264	2.415	2.624	2.771	2.977	3.326	4.140
15	2.249	2.397	2.602	2.746	2.947	3.286	4.073
16	2.235	2.382	2.583	2.724	2.921	3.252	4.015
17	2.224	2.368	2.567	2.706	2.898	3.222	3.965
18	2.214	2.356	2.552	2.689	2.878	3.197	3.922
19	2.205	2.346	2.539	2.674	2.861	3.174	3.883
20	2.197	2.336	2.528	2.661	2.845	3.153	3.849
21	2.189	2.328	2.518	2.649	2.831	3.135	3.819
22	2.183	2.320	2.508	2.639	2.819	3.119	3.792
23	2.177	2.313	2.500	2.629	2.807	3.104	3.768
24	2.172	2.307	2.492	2.620	2.797	3.091	3.745
25	2.167	2.301	2.485	2.612	2.787	3.078	3.725
26	2.162	2.296	2.479	2.605	2.779	3.067	3.707
27	2.158	2.291	2.473	2.598	2.771	3.057	3.690
28	2.154	2.286	2.467	2.592	2.763	3.047	3.674
29	2.150	2.282	2.462	2.586	2.756	3.038	3.659
30	2.147	2.278	2.457	2.581	2.750	3.030	3.646
40	2.125	2.250	2.423	2.542	2.704	2.971	3.551
60	2.099	2.223	2.390	2.504	2.660	2.915	3.460
120	2.076	2.196	2.358	2.468	2.617	2.860	3.373
∞	2.054	2.170	2.326	2.432	2.576	2.807	3.291

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Laplace Transform Table

		$f(t)$	$F(s)$
1.	Unit impulse	$\delta(t)$	1
2.	Unit step	$u(t)$	$\frac{1}{s}$
3.	Unit ramp	$r(t) = tu(t)$	$\frac{1}{s^2}$
4.	Unit parabola	$p(t) = \frac{1}{2}t^2u(t)$	$\frac{1}{s^3}$
5.	Exponential	e^{-at}	$\frac{1}{s+a}$
6.	t-Multiplication exponential	te^{-at}	$\frac{1}{(s+a)^2}$
7.	Sine	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
8.	Cosine	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
9.	Damped Sine	$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
10.	Damped Cosine	$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
11.	Linearity	$c_1f_1(t)+c_2f_2(t)$	$c_1F_1(s)+c_2F_2(s)$
12.	Differentiation	$\frac{d}{dt}f(t)$	$sF(s)-f(0)$
13.	n-Fold differentiation	$\frac{d^n}{dt^n}f(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)$ $-\cdots\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$
14.	Integration	$\int_{0-}^{t} f(\tau)d\tau$	$\frac{F(s)}{s}$
15.	t-Multiplication	$tf(t)$	$\frac{-d}{ds}F(s)$

Note: Laplace Transform Table continues on page 13.

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Laplace Transform Table (Continued)

	$f(t)$	$F(s)$
16. n-Fold t-Multiplication	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$
17. Time shift	$f(t-t_0)u(t-t_0)$; $t_0 > 0$	$e^{-st_0} F(s)$
18. Frequency shift	$e^{-s_0 t} f(t)$	$F(s+s_0)$
19. Time-frequency scaling	$f(ct)$; $c > 0$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
20. Periodic Function	$f(t) = f_1(t)u(t) + f_1(t-T)u(t-T) + f_1(t-2T)u(t-2T) + \cdots$	$\frac{1}{1-e^{-sT}} F_1(s)$

END OF PAPER

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IE2106 ENGINEERING MATHEMATICS I

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